

GNN Final Report

Target: BACC > 0.94

Phase 1: Optimization (5-Fold CV)

Best Average CV Score: 0.9411

(This score determined the best parameters below)

Best Hyperparameters Found:

model_type: GAT
num_layers: 5
hidden_channels: 64
heads: 4
dropout: 0.23
lr: 3.76e-04
batch_size: 64
weight_decay: 9.01e-04

Phase 2: External Validation (Final Check)

Tested on 'moleculas_unicas.csv' (never seen during training)

External BACC (Thr 0.5): 0.8911
External BACC (Optimal): 0.9371 (Thr=0.19)
Target Met (>0.94): NO ☐

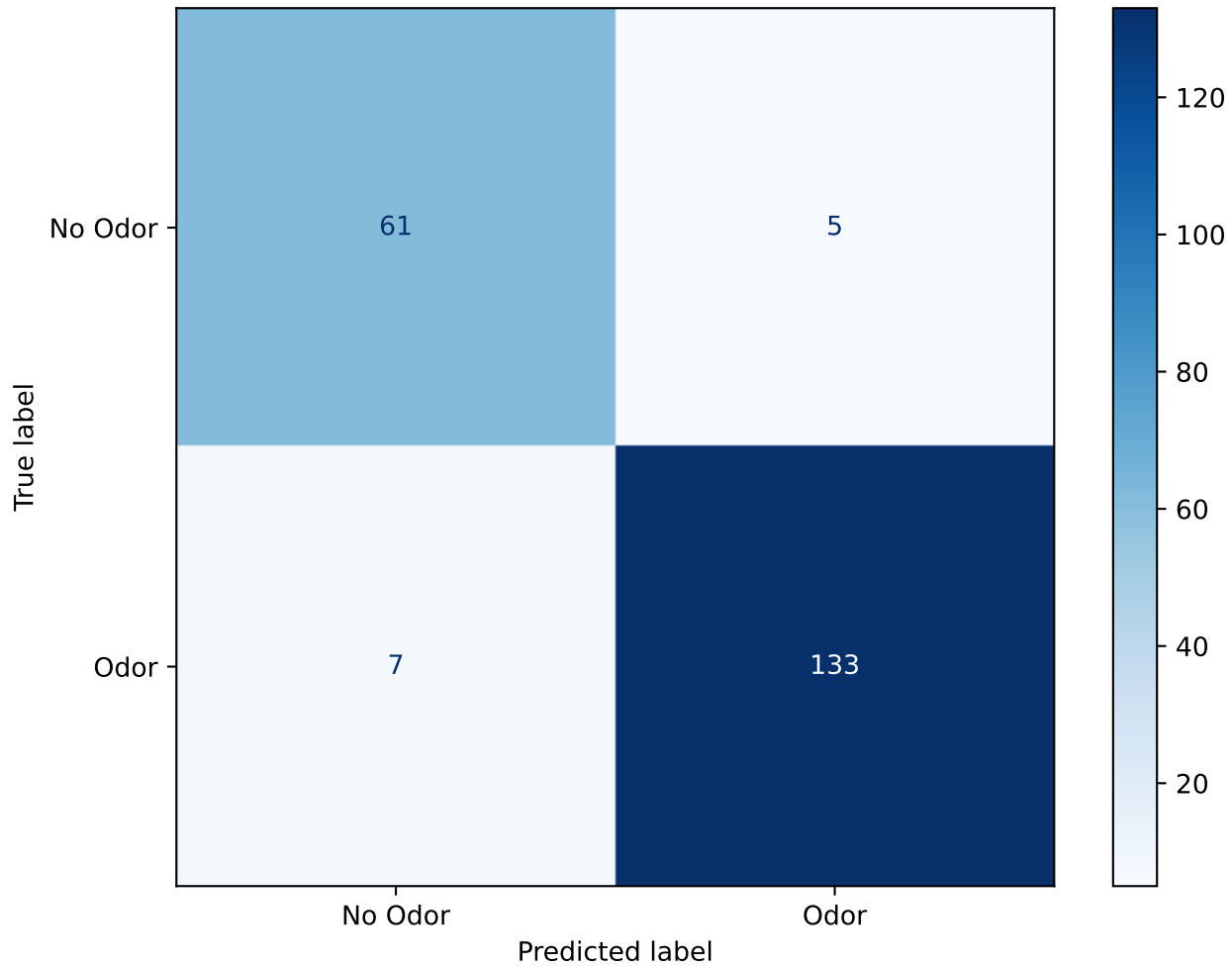
Detailed External Metrics:

[Threshold 0.5]	[Optimal Thr 0.19]
MCC: 0.74	MCC: 0.87
Precision: 0.97	Precision: 0.96
Recall: 0.84	Recall: 0.95
F1 Score: 0.90	F1 Score: 0.96

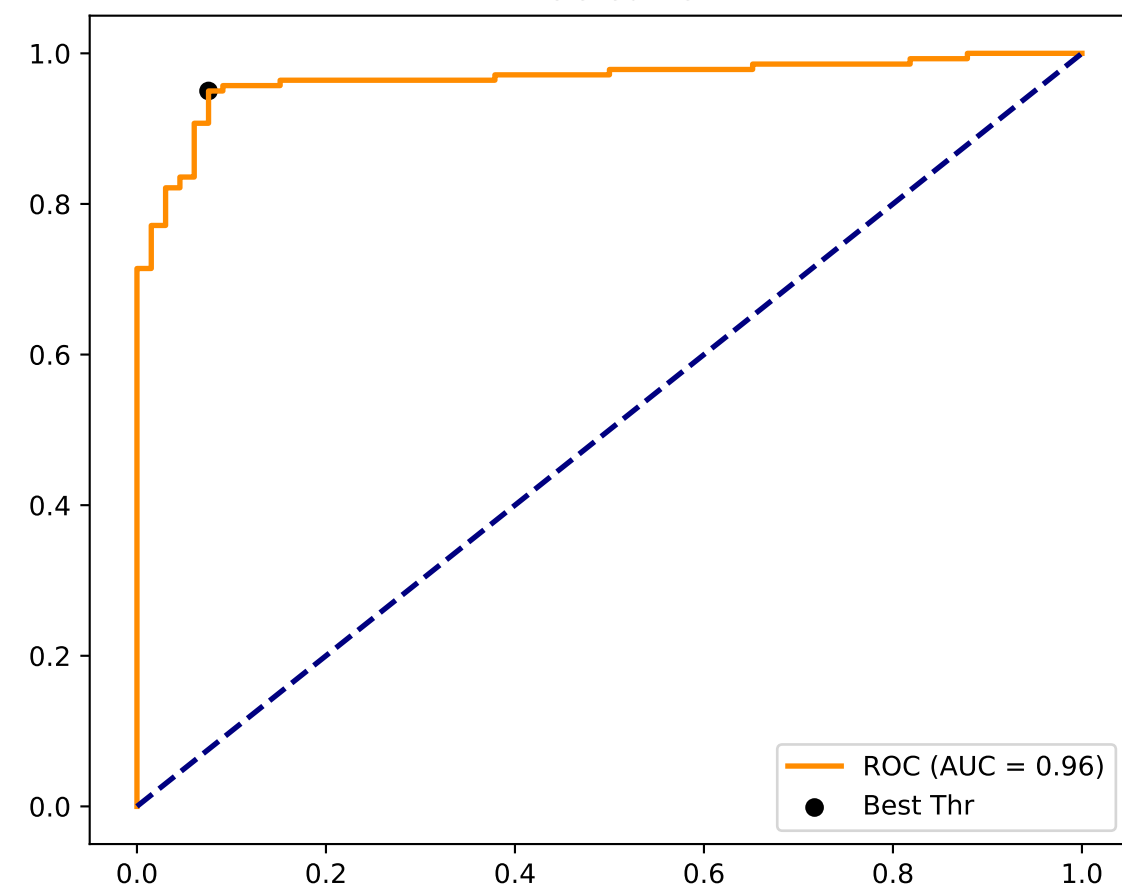
Dataset Statistics:

Train Set: 4011 (Pos:3398 Neg:613)
Ext Set: 206 (Pos:140 Neg:66)

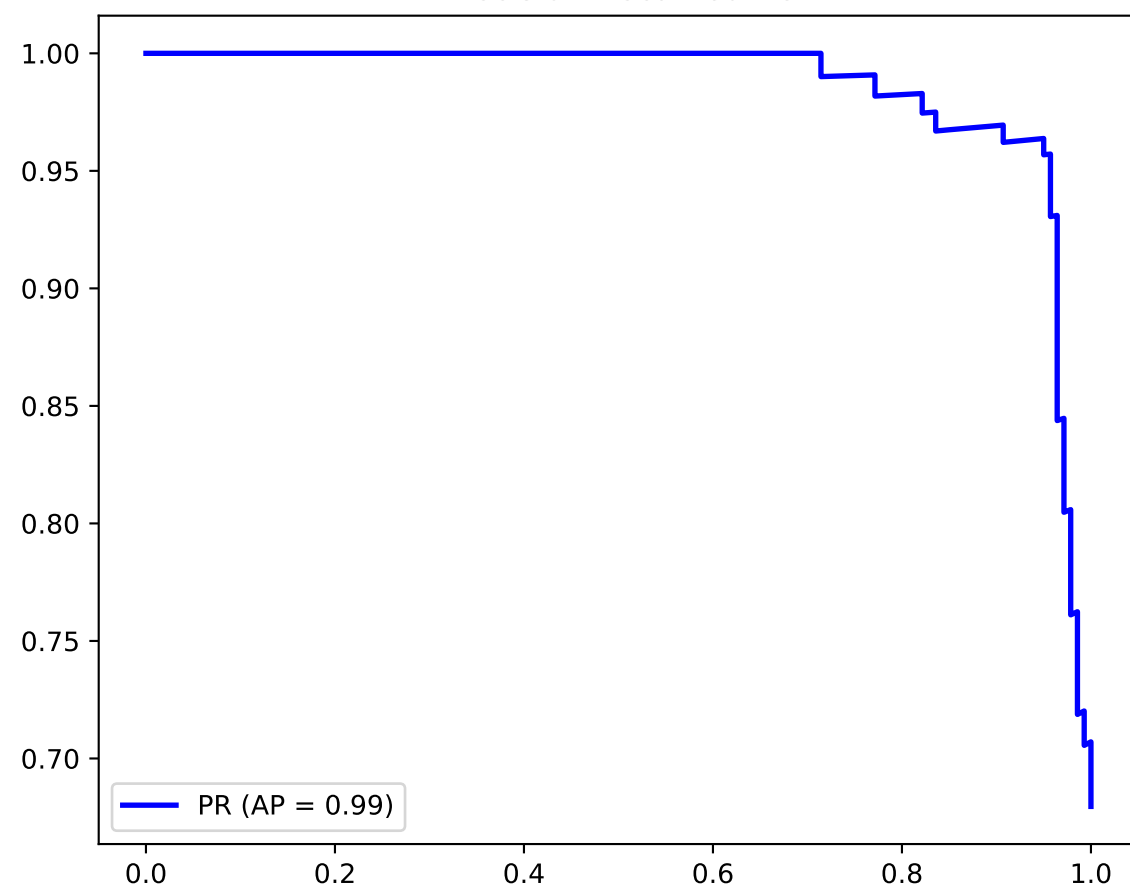
Confusion Matrix (Optimal Threshold 0.19)



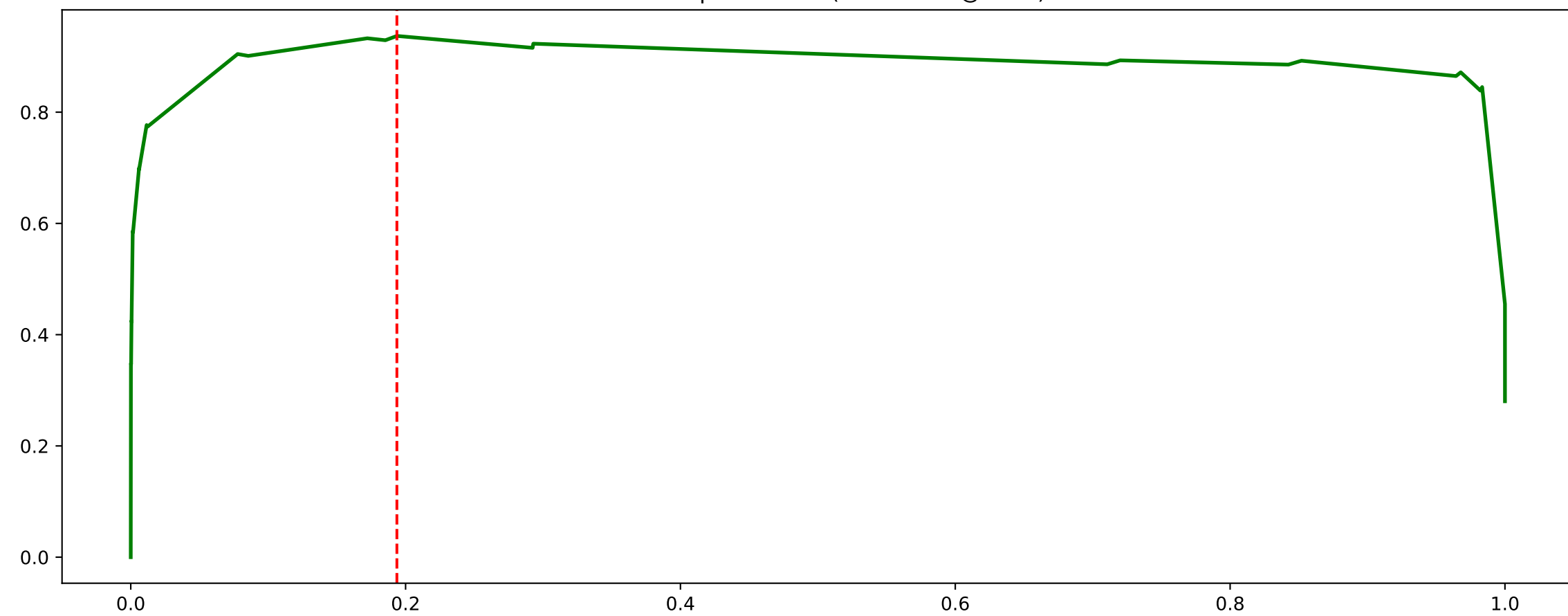
ROC Curve



Precision-Recall Curve



G-Mean Optimization (Best: 0.94 @ 0.19)



GNN Architecture Diagram

